# CSA1613-DATAWAREHOUSING AND DATAMINING
# LAB EXPERIMENTS
# R PROGRAMMING

## EXPERIMENT-1 ADDITION:

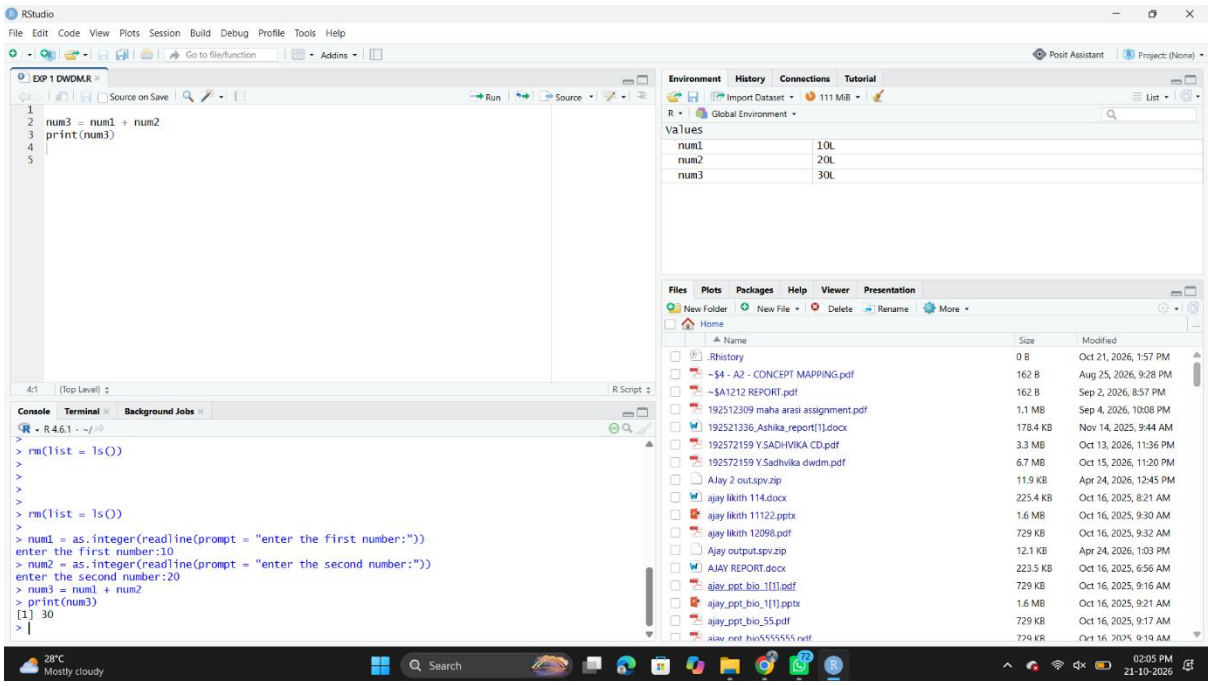
AIM:

To prove the program for addition using R-tool.

PROGRAM:

```
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
num3=num1+num2
print(num3)
```

OUTPUT:




RESULT:

Thus the basic program addition is executed successfully.

## EXPERIMENT-2 SUBTRACTION:

AIM:

To prove the program for subtraction using R-tool.

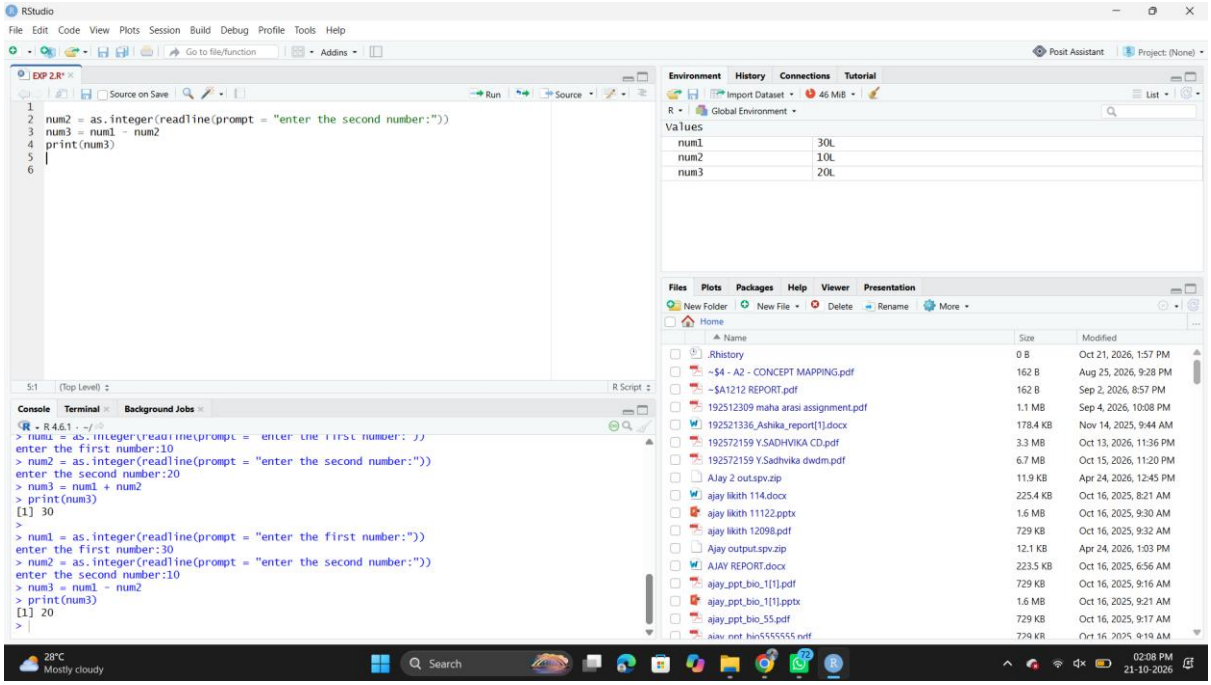
PROGRAM:

```
num1=as.integer(readline(prompt = "enter the first number:"))
num2=as.integer(readline(prompt = "enter the second number:"))
```

num3=num1-num2

print(num3)

OUTPUT:




RESULT:

Thus, the basic program subtraction is executed successfully.

EXPERIMENT-3 MULTIPLICATION:

AIM:

To prove the program for multiplication using R-tool.

PROGRAM:

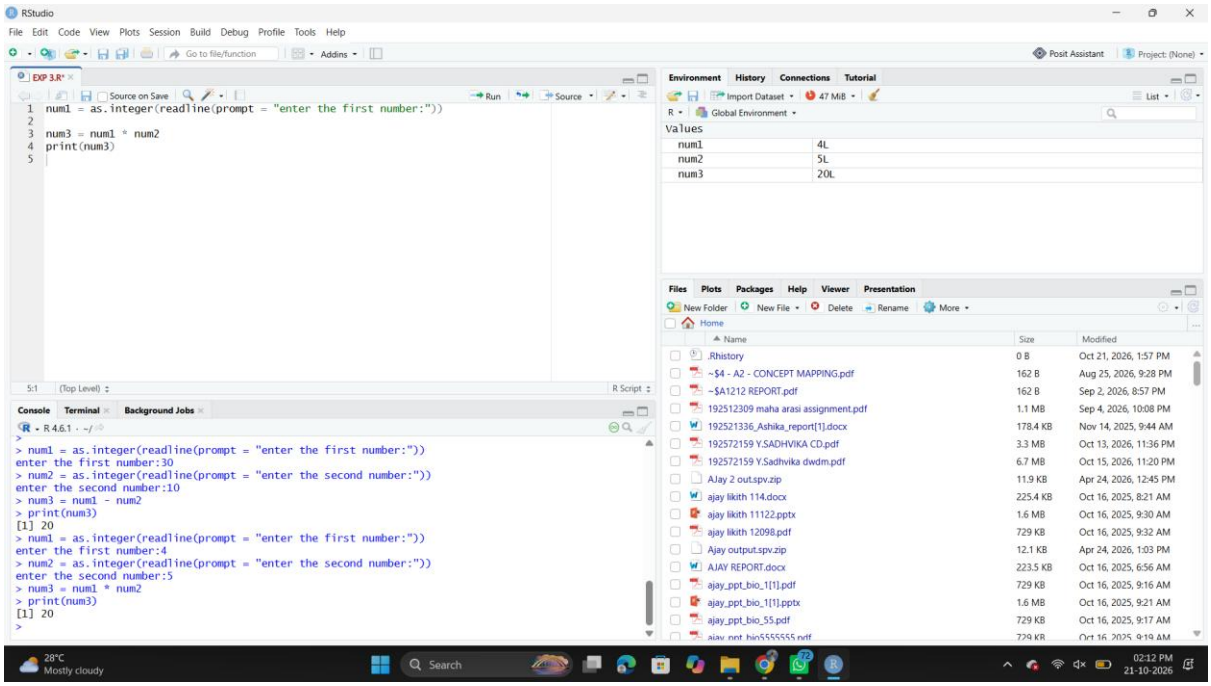
num1=as.integer(readline(prompt = "enter the first number:"))

num2=as.integer(readline(prompt = "enter the second number:"))

num3=num1*num2

print(num3)

OUTPUT:




RESULT:

Thus, the basic program multiplication is executed successfully.

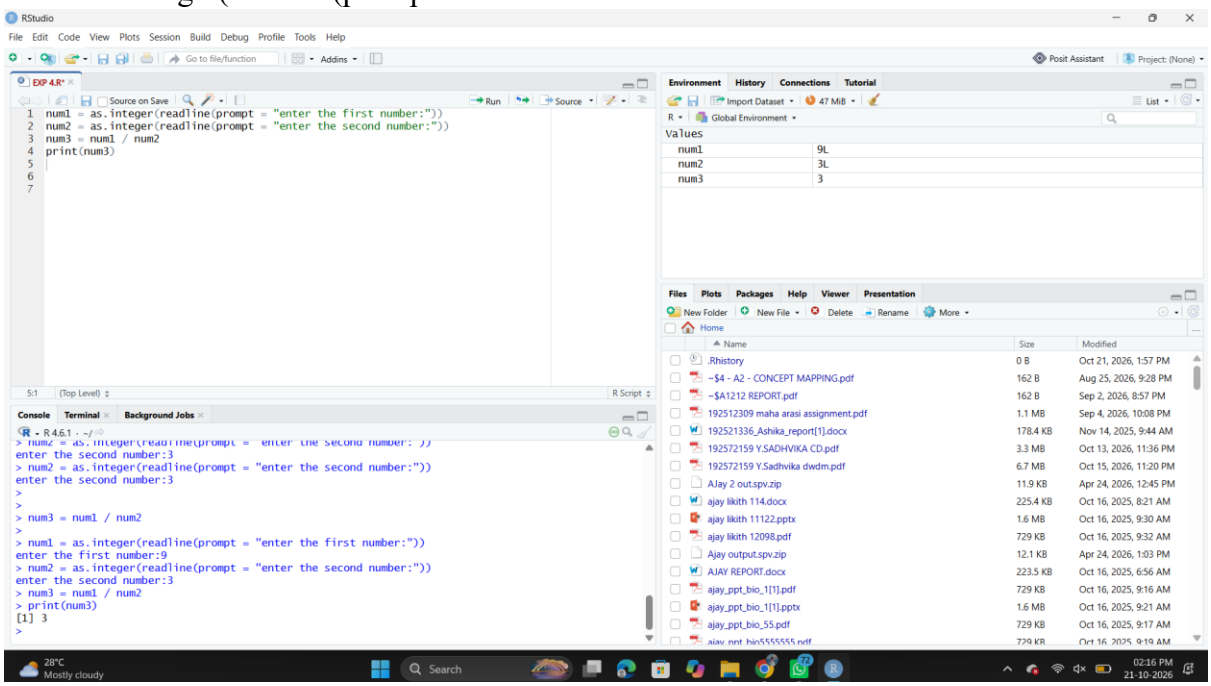
EXPERIMENT-4 DIVISION:

AIM:

To prove the program for division using R-tool.

PROGRAM:

num1=as.integer(readline(prompt =




"enter the first number:"))

num2=as.integer(readline(prompt = "enter the second number:"))

num3=num1/num2

print(num3)

OUTPUT:

RESULT:

Thus, the basic program division was executed successfully.

EXPERIMENT-5 ODD OR EVEN:

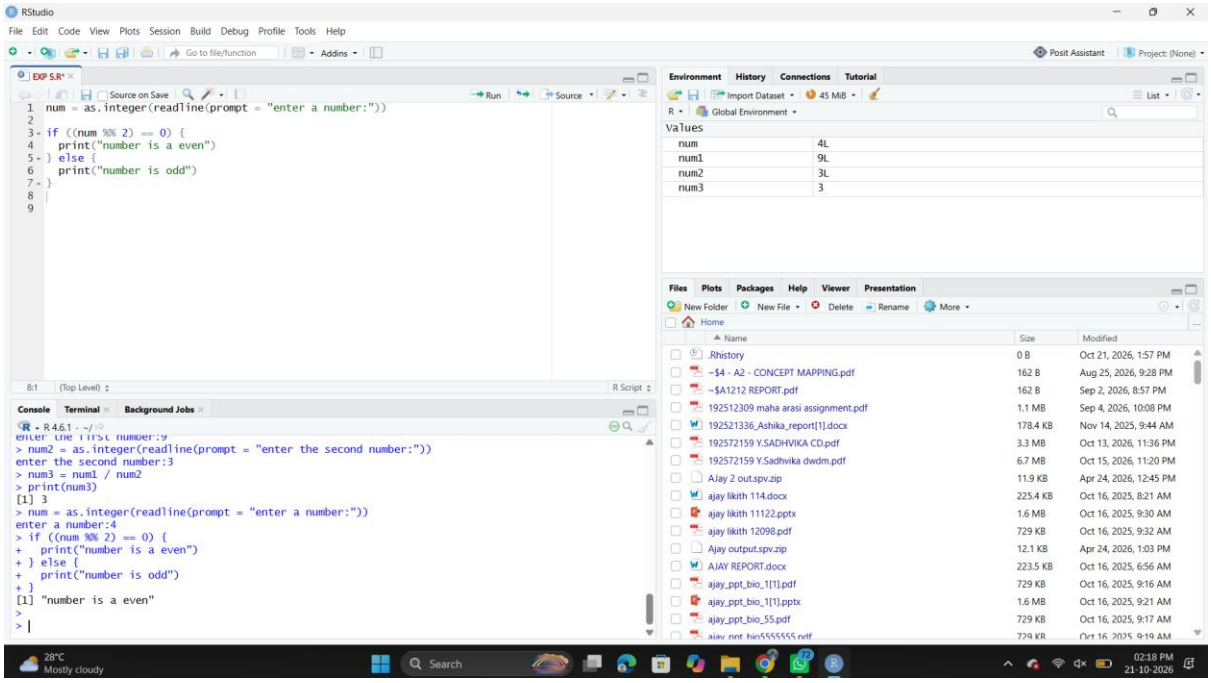
AIM:

To write the program for odd or even using R-tool.

PROGRAM:

```
num=as.integer(readline(prompt="enter a number:"))
if((num%%2)==0)
{
  print("number is a even")
}else{
  print("number is odd")
}
```

OUTPUT:




RESULT:

Thus, the basic program odd or even was executed successfully.

EXPERIMENT-6 MEAN,MEDIAN,MODE:

AIM:

To write the program for mean,median,mode.

PROGRAM:

MEAN

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
mean(df $age)
write.csv(df,"datafr.csv")
```

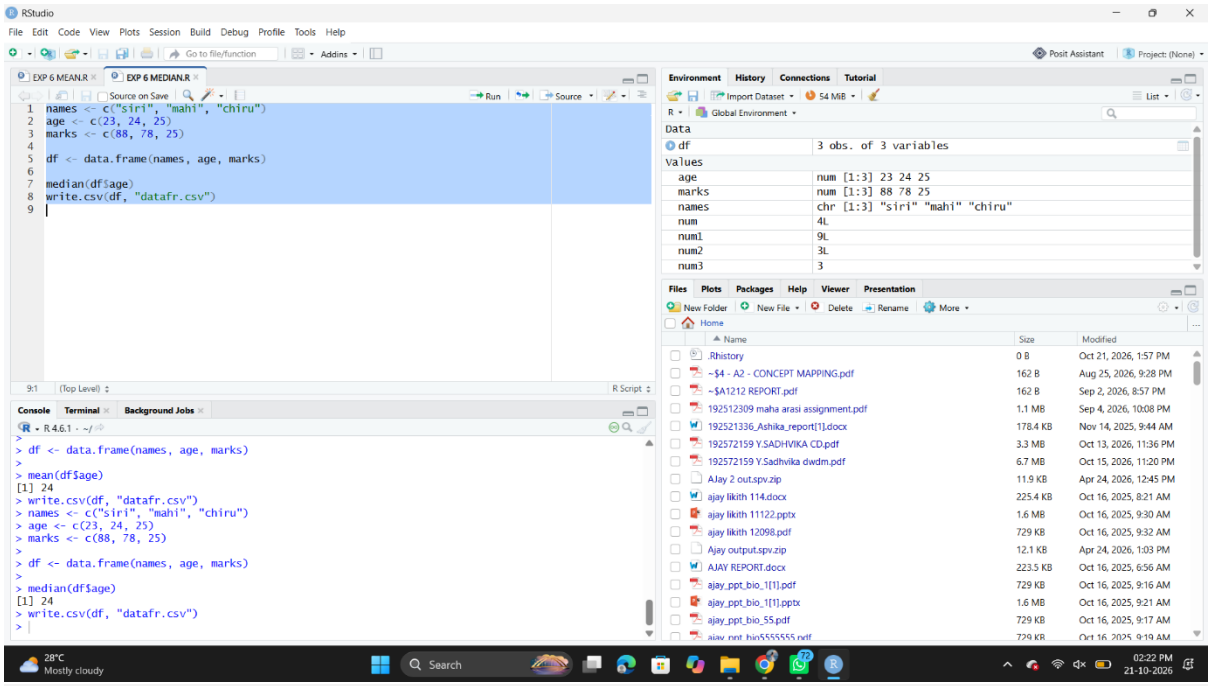
MEDIAN

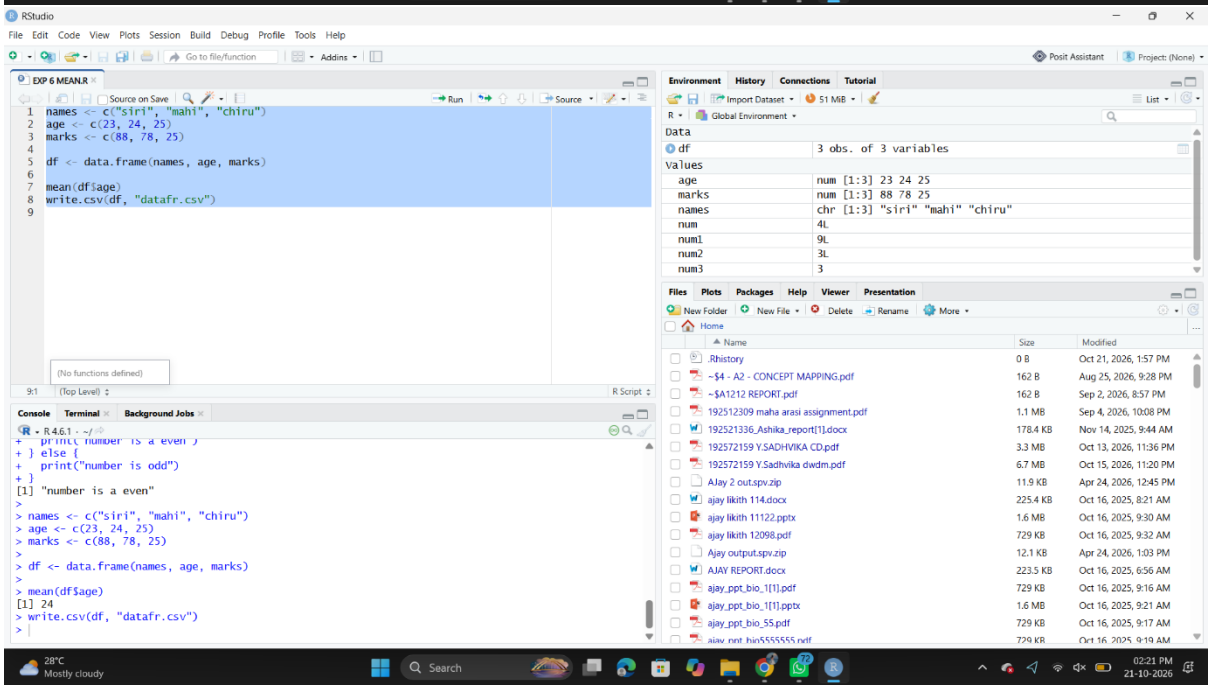
```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
median(df $age)
write.csv(df,"datafr.csv")
```

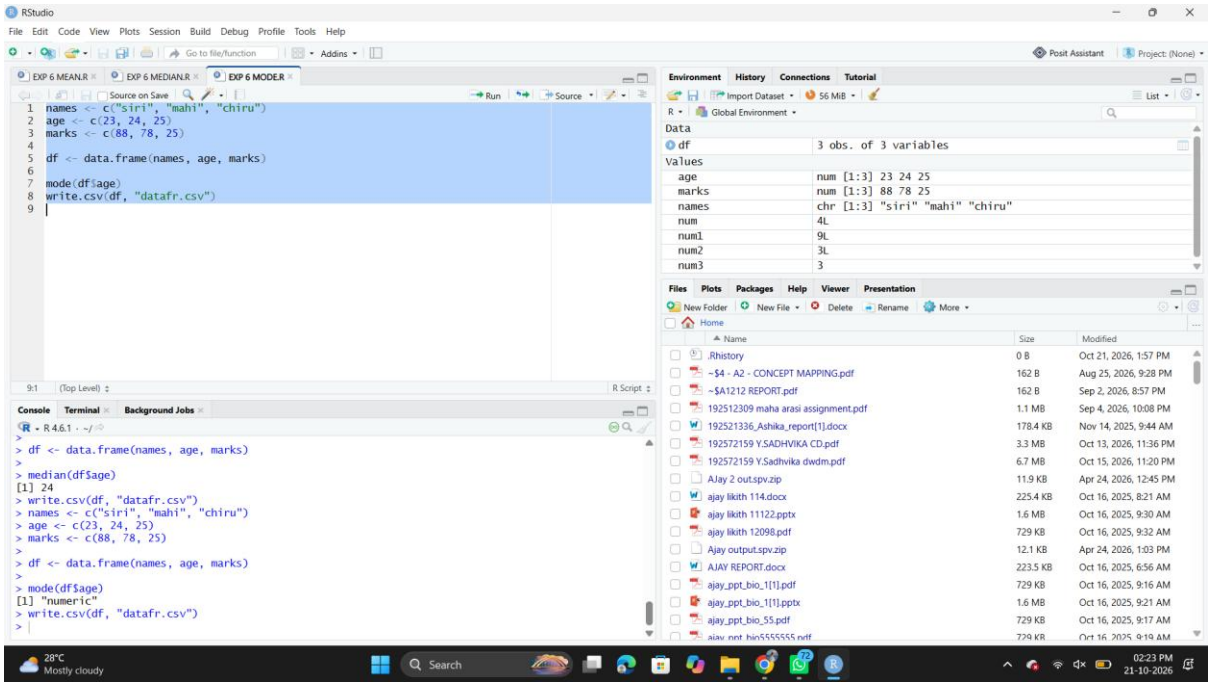
MODE

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
mode(df $age)
write.csv(df,"datafr.csv")
```

OUTPUT:


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
EXP 6 MEAN.R
EXP 6 MEDIAN.R
1 names <- c("siri", "mahi", "chiru")
2 age <- c(23, 24, 25)
3 marks <- c(88, 78, 25)
4
5 df <- data.frame(names, age, marks)
6
7 median(df$age)
8 write.csv(df, "datafr.csv")
9
Environment History Connections Tutorial
Data
df 3 obs. of 3 variables
Values
age num [1:3] 23 24 25
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 4L
num1 9L
num2 3L
num3 3
Files Plots Packages Help Viewer Presentation
Console Terminal Background Jobs
> df <- data.frame(names, age, marks)
> mean(df$age)
[1] 24
> write.csv(df, "datafr.csv")
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
> median(df$age)
[1] 24
> write.csv(df, "datafr.csv")
28°C Mostly cloudy
Search
02:22 PM 21-10-2026


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
EXP 6 MEAN.R
1 names <- c("siri", "mahi", "chiru")
2 age <- c(23, 24, 25)
3 marks <- c(88, 78, 25)
4
5 df <- data.frame(names, age, marks)
6
7 mean(df$age)
8 write.csv(df, "datafr.csv")
9
(No functions defined)
Environment History Connections Tutorial
Data
df 3 obs. of 3 variables
Values
age num [1:3] 23 24 25
marks num [1:3] 88 78 25
names chr [1:3] "siri" "mahi" "chiru"
num 4L
num1 9L
num2 3L
num3 3
Files Plots Packages Help Viewer Presentation
Console Terminal Background Jobs
+ print("number is a even")
+ } else {
+ print("number is odd")
+ }
[1] "number is a even"
> names <- c("siri", "mahi", "chiru")
> age <- c(23, 24, 25)
> marks <- c(88, 78, 25)
> df <- data.frame(names, age, marks)
> mean(df$age)
[1] 24
> write.csv(df, "datafr.csv")
28°C Mostly cloudy
Search
02:21 PM 21-10-2026




RESULT:

Thus the central tendency and measure of dispersion is executed successfully.

EXPERIMENT-7 SUMMARY:

AIM:

To write the program for summary using R-tool.

PROGRAM:

names<-c("siri","mahi","chiru")

age<-c(23,24,25)

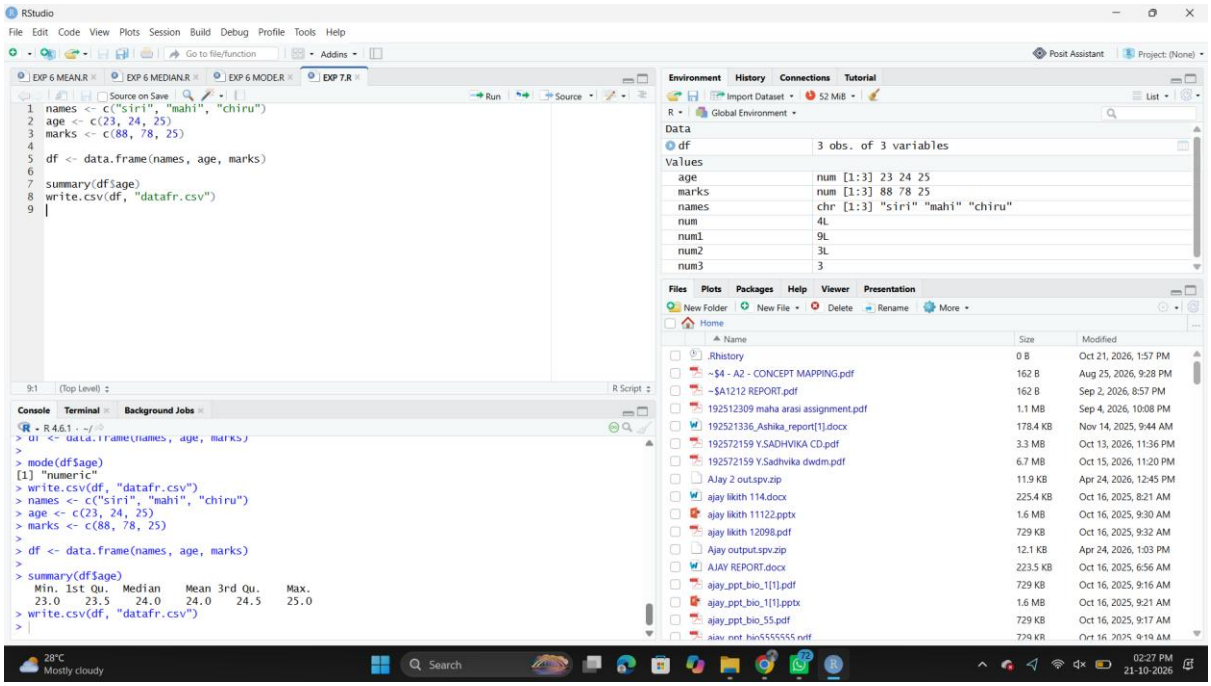
marks<-c(88,78,25)

df<-data.frame(names,age,marks)

summary(df $age)

write.csv(df,"datafr.csv")

OUTPUT:




RESULT:

Thus, the central tendancy and measure of dispersion is executed successfully.

EXPERIMENT-8 GREATER AMONG THREE NUMBERS:

AIM:

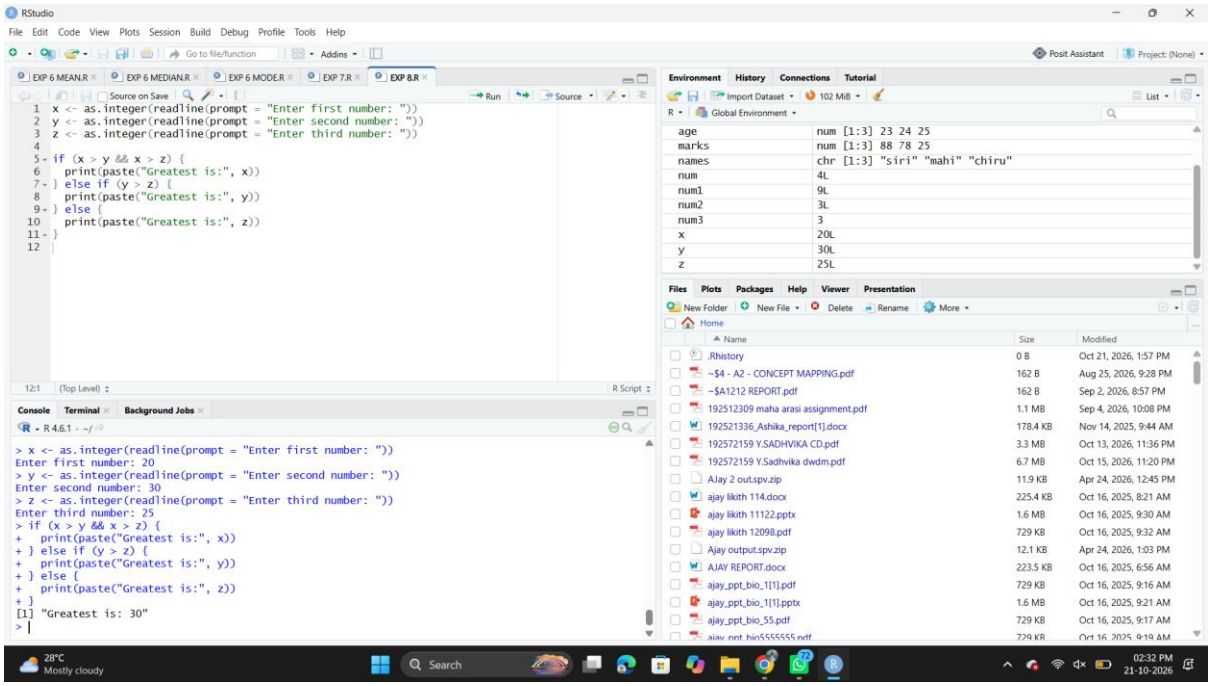
To write the program for the greatest among three numbers.

PROGRAM:

```
x <- as.integer(readline(prompt = "Enter first number :"))
y <- as.integer(readline(prompt = "Enter second number :"))
z <- as.integer(readline(prompt = "Enter third number :"))

if (x > y && x > z) {
  print(paste("Greatest is :", x))
} else if (y > z) {
  print(paste("Greatest is :", y))
} else{
  print(paste("Greatest is :", z))
}
```

OUTPUT:




RESULT:

Thus, the greatest among the three numbers was executed successfully.

EXPERIMENT-9 IQR:

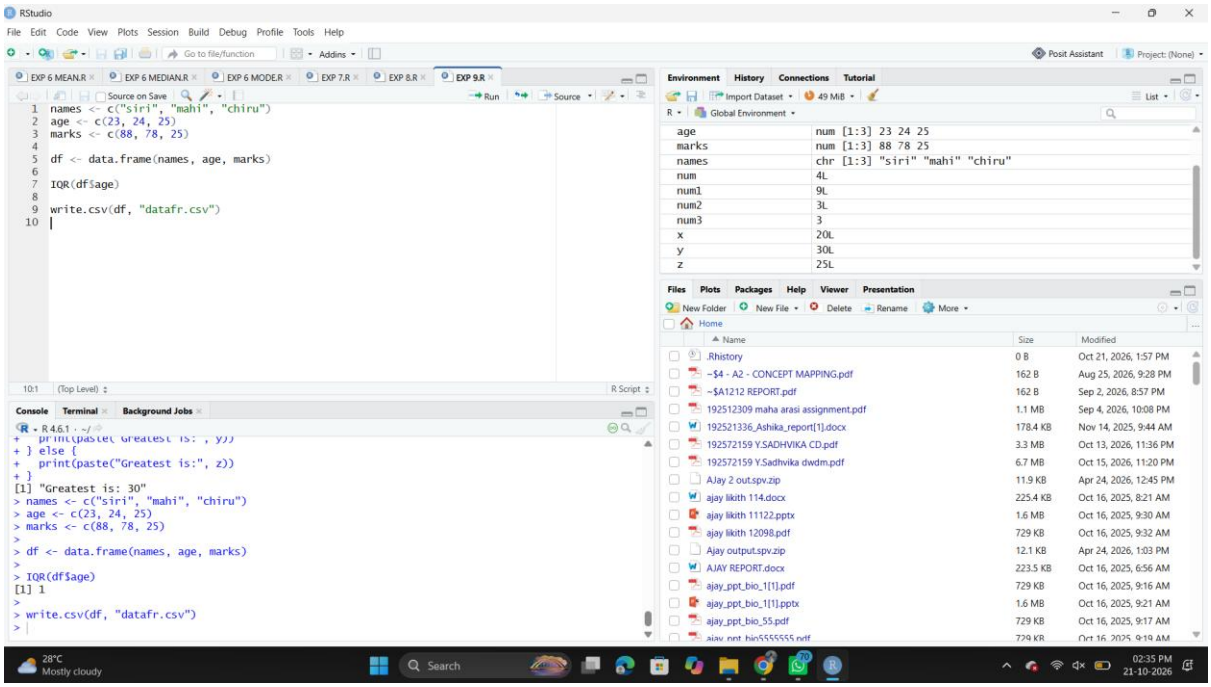
AIM:

To write the program for central tendency and data dispersion measures using R tool.

PROGRAM:

```
names<-c("siri","mahi","chiru")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
IQR(df $age)
write.csv(df,"datafr.csv")
```

OUTPUT:




RESULT:

Thus, the program for central tendency and data dispersion measures was executed successfully.

EXPERIMENT-10 QUANTILE:

AIM:

To write the program for central tendency and data dispersion measures.

PROGRAM:

names<-c("siri","mahi","chiru")

age<-c(23,24,25)

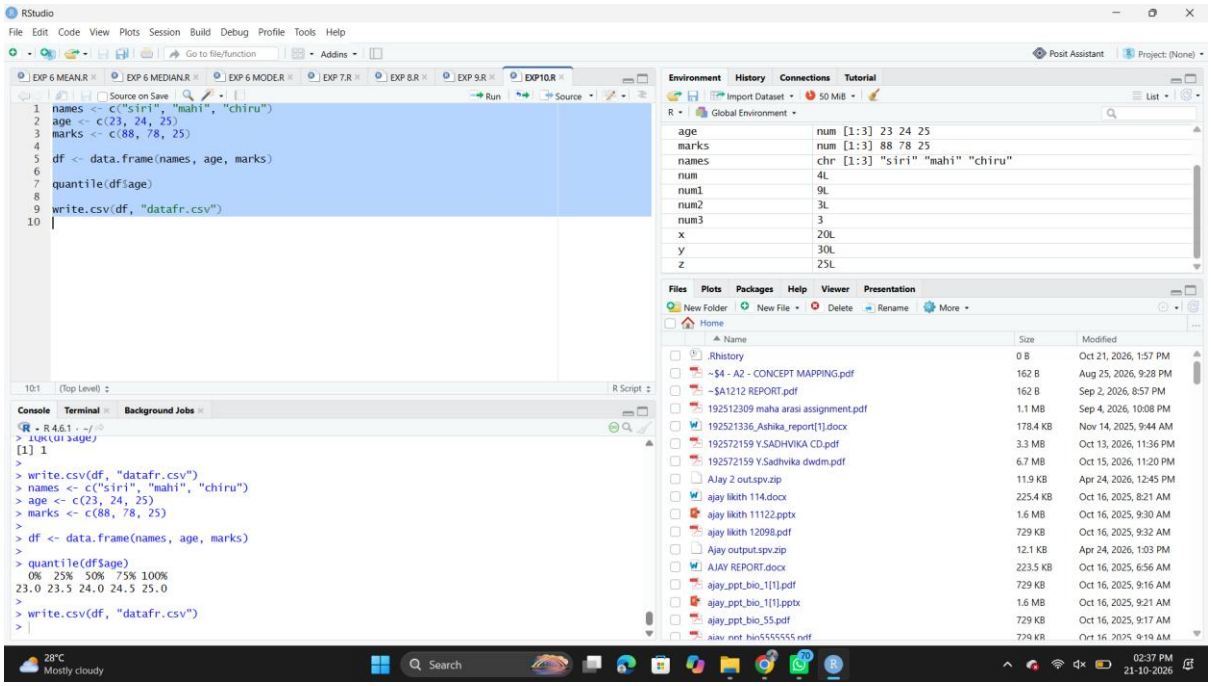
marks<-c(88,78,25)

df<-data.frame(names,age,marks)

quantile(df $age)

write.csv(df,"datafr.csv")

OUTPUT:




RESULT:

Thus, the program for central tendency and data dispersion measures was executed successfully

EXPERIMENT-11MID RANGE:

AIM:

To write the program for central tendency and data dispersion measures.

PROGRAM:

names<-c("siri","mahi","chiru")

age<-c(23,24,25)

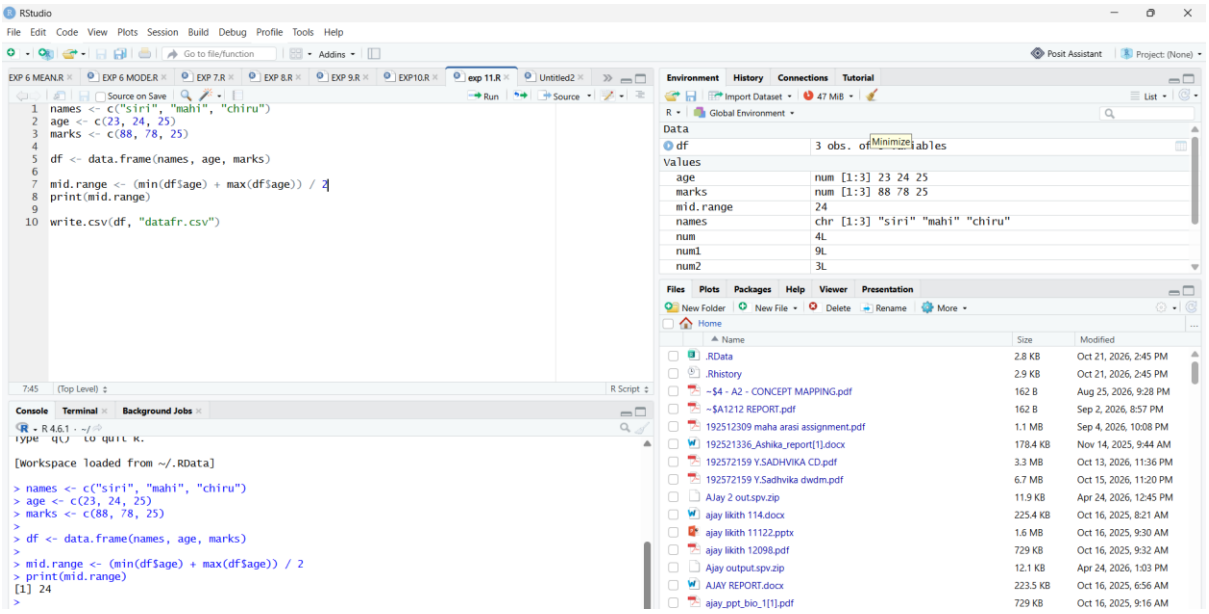
marks<-c(88,78,25)

df<-data.frame(names,age,marks)

mid range(df $age)

write.csv(df,"datafr.csv")

OUTPUT:




RESULT:

Thus, the program for central tendency and data dispersion measures was executed successfully

EXPERIMENT-12 Z-SCOORE NORMALIZATION:

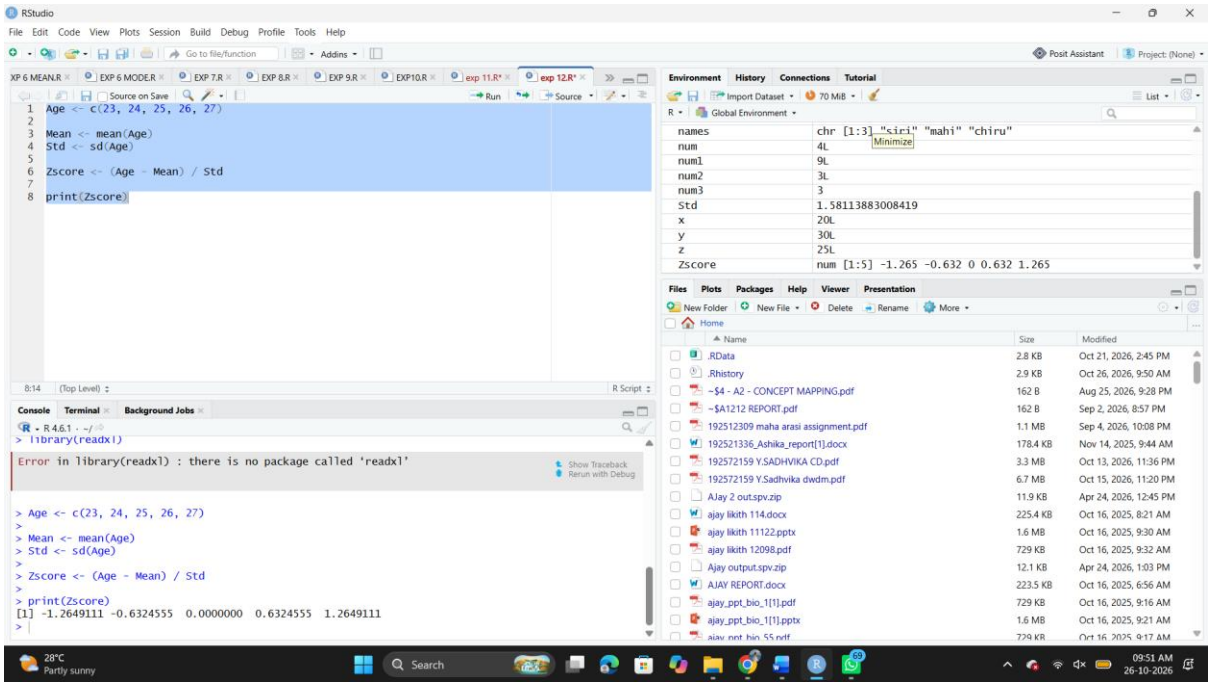
AIM:

To write the program for Z-scoore normalization using R-tool.

PROGRAM:

```
diabetest1<-read_excel("C:/Users/Krisha/Downloads/NARA.xlsx")
A<-c(diabetest1$Age)
Mean<-mean(A)
Std<-sd(A)
Zscore<-(A-Mean)/Std
Zscore
```

OUTPUT:




RESULT:
Thus, the Z-score normalization using R tool was executed successfully.

EXPERIMENT-13 MIN,MAX,MEAN,MINMAX:
AIM:

To write the program for the minimum,maximum,mean and minmax using r-TOOL

PROGRAM:

MEAN

```
diabetest1<-read_excel("C:/Users/Krisha/Downloads/NARA.xlsx")

A<-c(diabetest1$Age)

Mean<-mean(A)
```

MINIMUM

```
diabetest1<-read_excel("C:/Users/Krisha/Downloads/NARA.xlsx")

A<-c(diabetest1$Age)

Minimum<-Min(diabetest1$Age)
```

MAXIMUM

```
diabetest1<-read_excel("C:/Users/Krisha/Downloads/NARA.xlsx")

A<-c(diabetest1$Age)

Maximum<-Max(diabetest1$Age)
```

MINMAX

```
diabetest1<-read_excel("C:/Users/Krisha/Downloads/NARA.xlsx")
```

A<-c(diabetest1$Age)

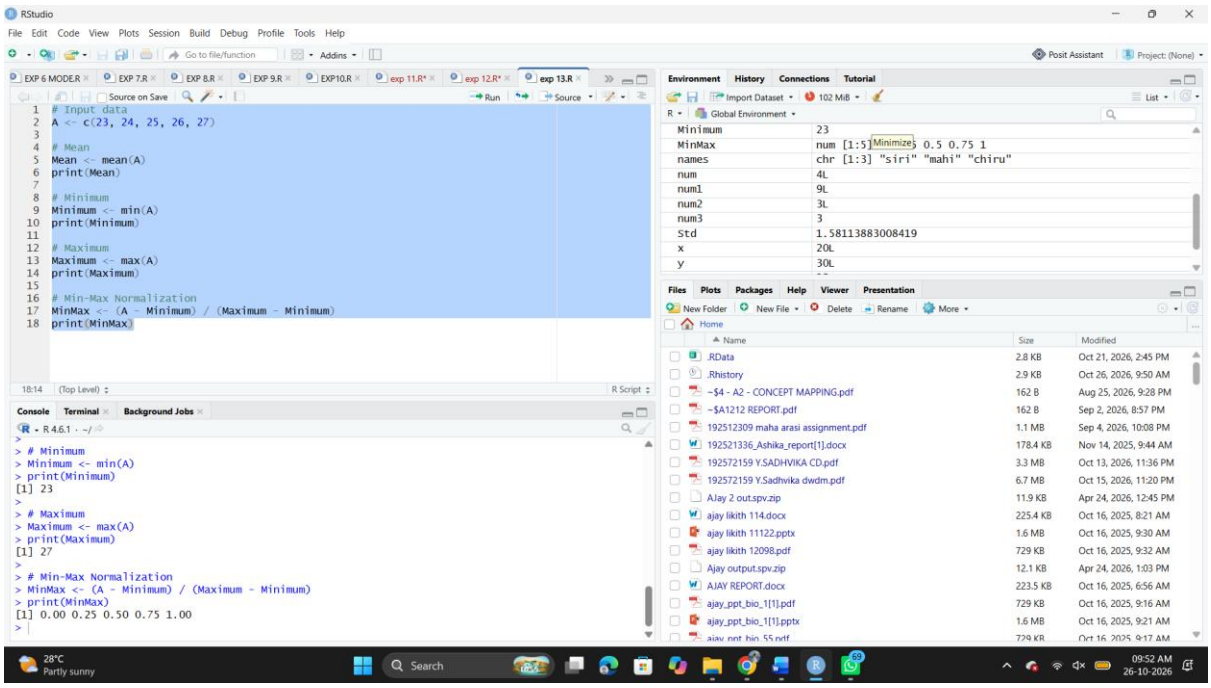
Maximum<-Max(diabetest1$Age)

Minimum<-Min(diabetest1$Age)

MinMax<-(A-Minimum)/(Maximum-Minimum)

MinMax

OUTPUT:




RESULT:

Thus, the program for min,max,minmax,mean was executed successfully.

EXPERIMENT-14

BAR PLOT AND HORIZONTAL BAR:

AIM:

To draw the bar plot and horizontal bar using R-tool.

PROGRAM:

a<-c(55,67,89,80,90)

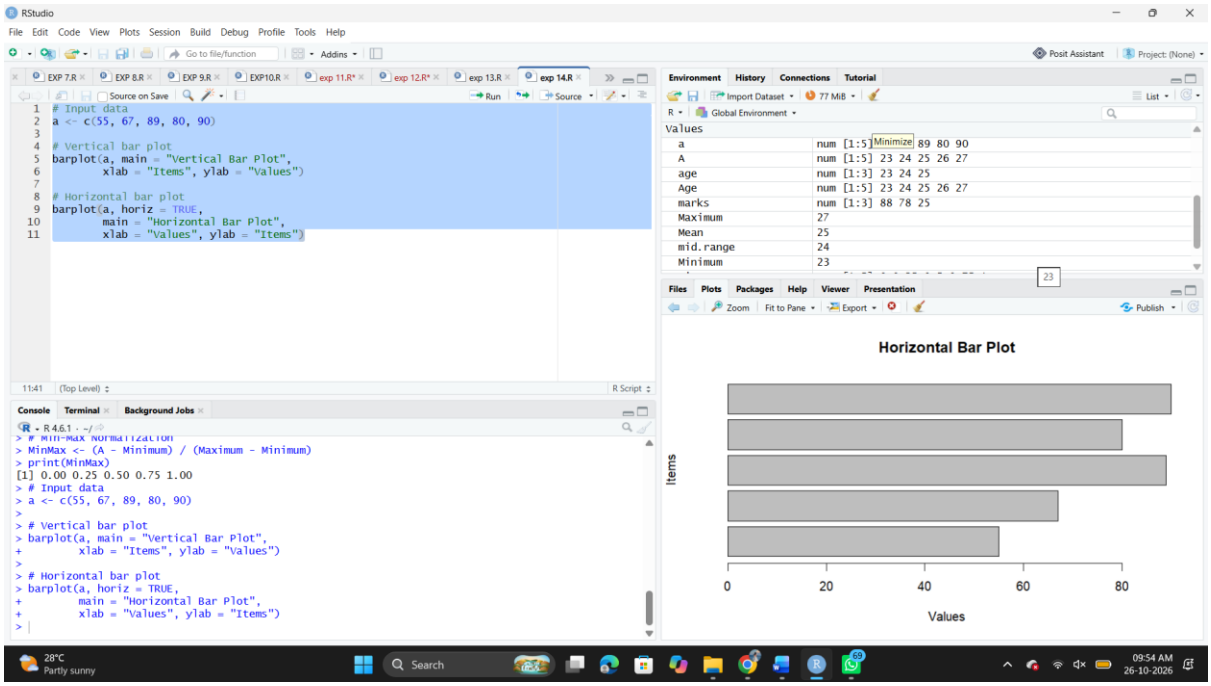
barplot(a)

a<-c(55,67,89,80,90)

barplot(a)

barplot(a,horiz=TRUE)

OUTPUT:




RESULT:

Thus, the bar and horizontal bar plot was executed successfully.

EXPERIMENT-15 BOX PLOT:

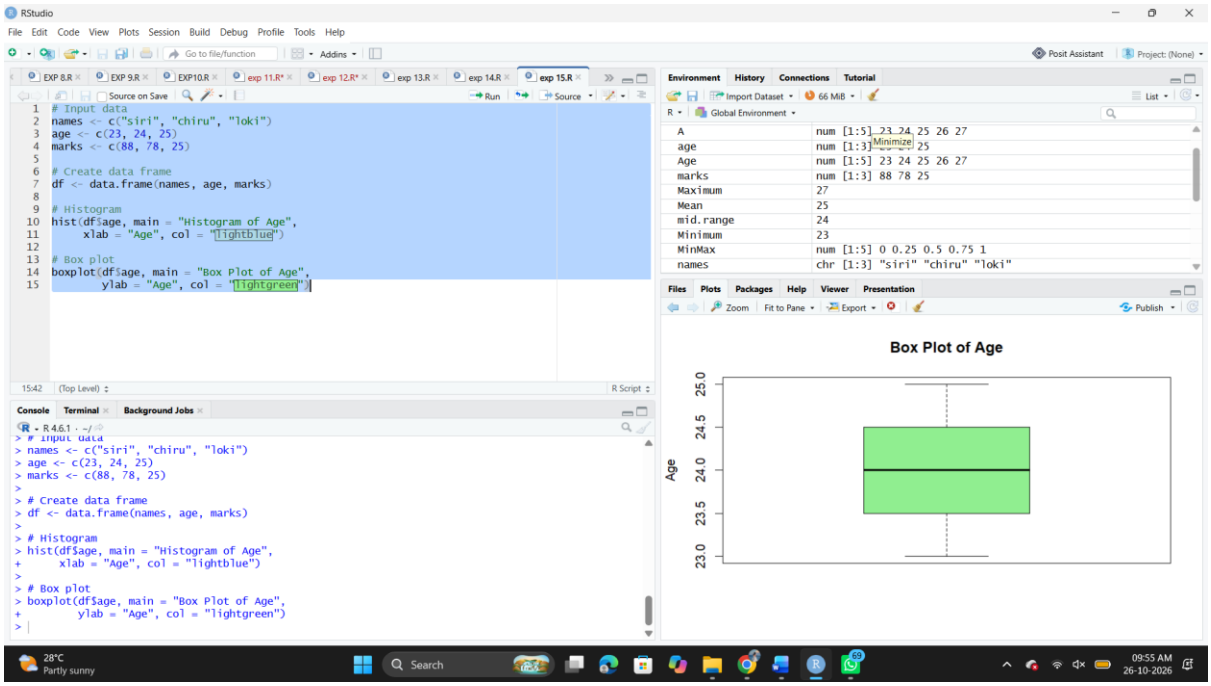
AIM:

To draw the box plot using R-tool.

PROGRAM:

```
names<-c("siri","chru","loki")
age<-c(23,24,25)
marks<-c(88,78,25)
df<-data.frame(names,age,marks)
hist(df$age)
boxplot(df$age)
```

OUTPUT:




RESULT:

Thus, the box plot was executed successfully.

EXPERIMENT-16 HISTOGRAM:

AIM:

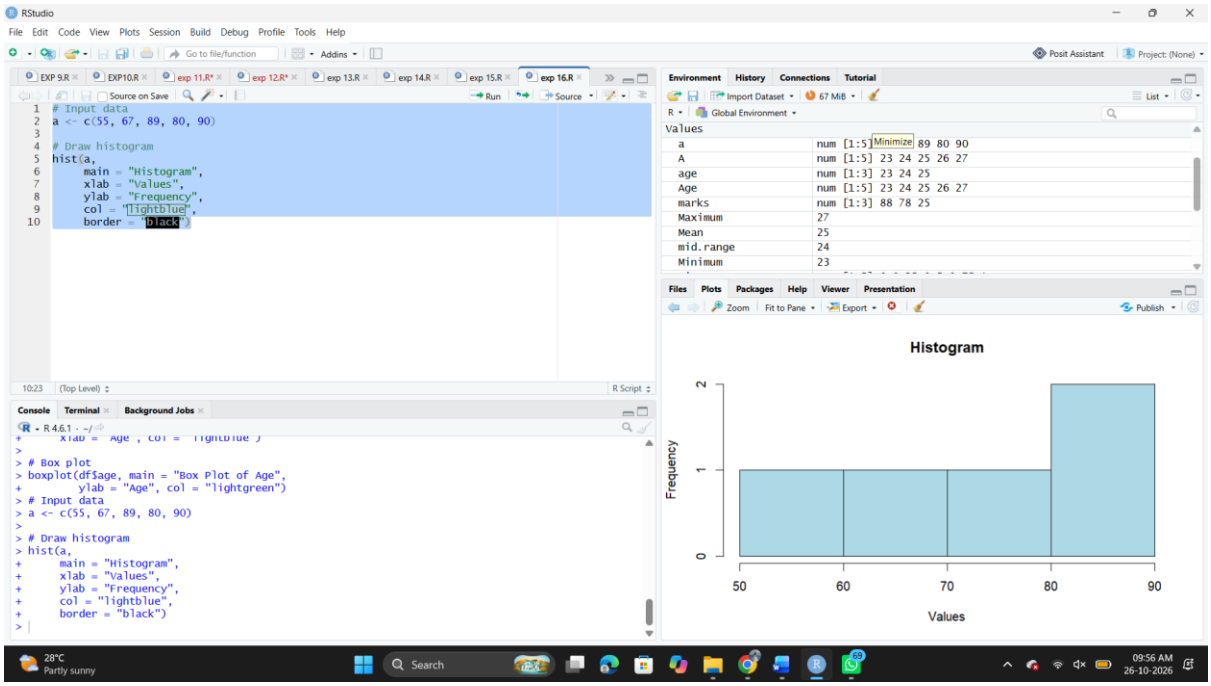
To draw the histogram plot using R-tooll.

PROGRAM:

a<-c(55,67,89,80,90)

hist(a)

OUTPUT:




RESULT:

Thus, the histogram plot was executed successfully.

EXPERIMENT-17 CORRELATION ANALYSIS:

AIM:

To write the program for correlation analysis using R-tool.

PROGRAM:

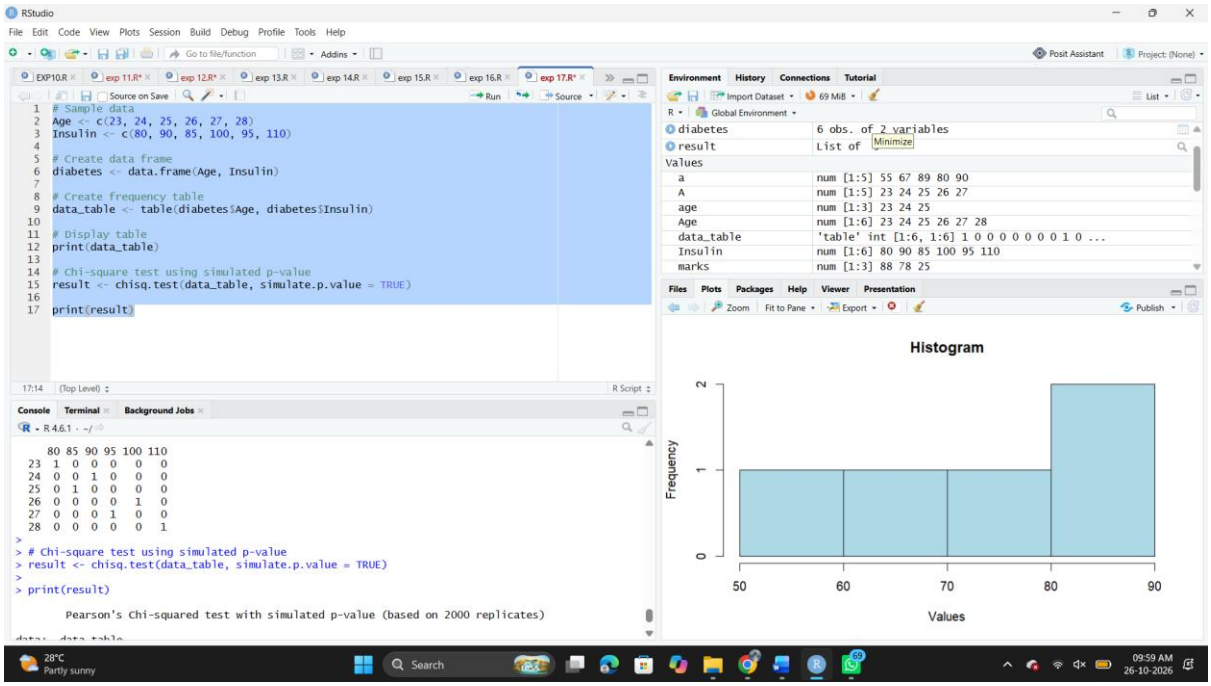
diabetest1<-read_excel("C:/Users/Krisha/Downloads/NARA.xlsx")

diabetest1<-table(diabetest1 $Age,diabetest1 $Insulin)

diabetest1

chisq.test(diabetest1)

OUTPUT:




RESULT:

Thus, the correlation analysis was executed successfully.

EXPERIMENT-18 SCATTER PLOT:

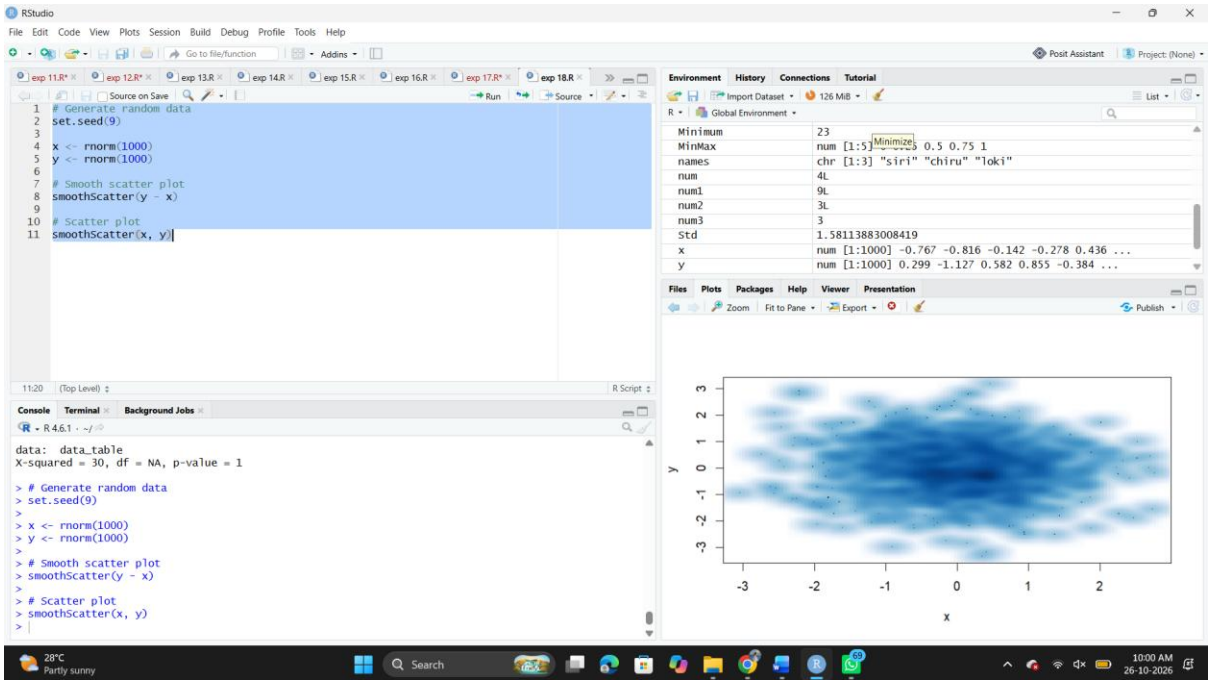
AIM:

To draw the scatter plot using R-tool

PROGRAM:

```
set.seed(9)
x <- rnorm(1000)
y <- rnorm(1000)
smoothScatter(y - x)
smoothScatter(x,y)
```

OUTPUT:




RESULT:

Thus, the scatter plot was executed successfully.

EXPERIMENT-19 LINEAR REGRESSION:

AIM:

To write thr program for the linear regression using R-tool.

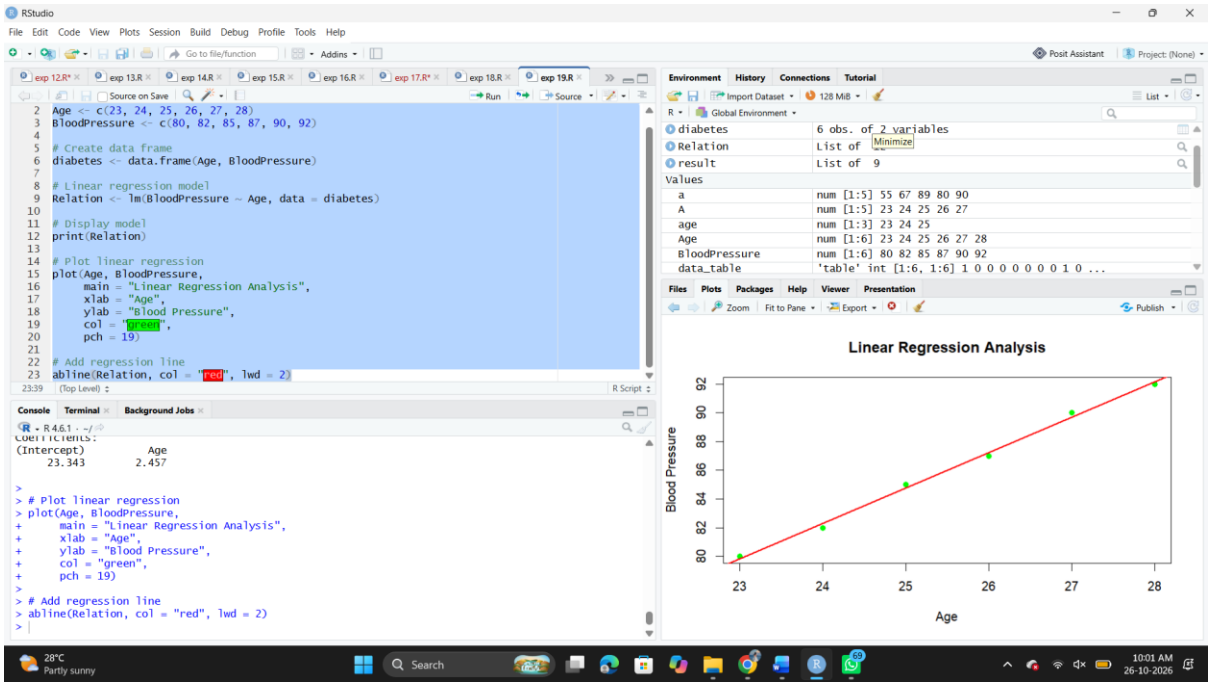
PROGRAM:

Relation <- lm(diabetes$BloodPressure~diabetes$Age)

Png<- (file=”linear regression.png”)

Plot(diabetes$Age, diabetes$BloodPressure, col=”green”, main= “ Linear Regression Analysis” , abline= (lm(diabetes$BloodPressure~ diabetes$Age)), xlab = “BloodPressure”, ylanb= “Age”)

OUTPUT:




RESULT:

Thus, the linear regression program was executed successfully.

EXPERIMENT-20

MULTIPLE REGRESSION:

AIM:

To write the program for the multiple regression.

PROGRAM:

Input <- diabetes[,c(“Age”, “BloodPressure”, “Glucose”)]

Model <- lm(Age~ BloodPressure+Glucose,data=input)

Print(model)

A<- coef(model)[1]

Print(A)

xBloodPressure<- coef(model)[2]

yGlucose<- coef(model)[3]

print(xBloodPressure)

print(yGlucose)

y = A+xBloodPressure + yGlucose

print(y)

**OUTPUT:**

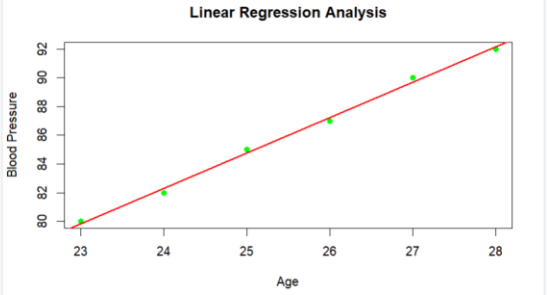
```
> # Generate random data
> set.seed(9)
>
> x <- rnorm(1000)
> y <- rnorm(1000)
>
> # Smooth scatter plot
> smoothScatter(y ~ x)
>
> # Scatter plot
> smoothScatter(x, y)
> # Input data
> Age <- c(23, 24, 25, 26, 27, 28)
> BloodPressure <- c(80, 82, 85, 87, 90, 92)
>
> # Create data frame
> diabetes <- data.frame(Age, BloodPressure)
>
> # Linear regression model
> Relation <- lm(BloodPressure ~ Age, data = diabetes)
>
> # Display model
> print(Relation)

Call:
lm(formula = BloodPressure ~ Age, data = diabetes)

Coefficients:
(Intercept)          Age
     23.343        2.457

>
> # Plot linear regression
> plot(Age, BloodPressure,
+      main = "Linear Regression Analysis",
+      xlab = "Age",
+      ylab = "Blood Pressure",
+      col = "green",
+      pch = 19)
>
```

| | |
|---|---|
| names | chr [1:3] "siri" "chiru" "loki" |
| num | 4L |
| num1 | 9L |
| num2 | 3L |
| num3 | 3 |
| Std | 1.58113883008419 |
| x | num [1:1000] -0.767 -0.816 -0.142 -0.278 0.436 ... |
| xBloodPressure | Named num 0.391 |
| y | num [1:1000] 0.299 -1.127 0.582 0.855 -0.384 ... |
| yGlucose | Named num 0.00451 |



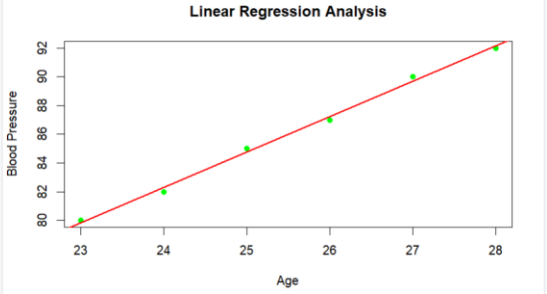

```
> # Add regression line
> abline(Relation, col = "red", lwd = 2)
> # Input data
> Age <- c(23, 24, 25, 26, 27, 28)
> BloodPressure <- c(80, 82, 85, 87, 90, 92)
> Glucose <- c(100, 110, 120, 125, 135, 140)
>
> # Create data frame
> diabetes <- data.frame(Age, BloodPressure, Glucose)
>
> # Multiple regression model
> model <- lm(Age ~ BloodPressure + Glucose,
+             data = diabetes)
>
> # Display model
> print(model)

Call:
lm(formula = Age ~ BloodPressure + Glucose, data = diabetes)

Coefficients:
  (Intercept)  BloodPressure        Glucose
    -8.672932       0.390977       0.004511

>
> # Display coefficients
> A <- coef(model)[1]
> print(A)
(Intercept)
  -8.672932
>
> xBloodPressure <- coef(model)[2]
> print(xBloodPressure)
BloodPressure
    0.3909774
>
> yGlucose <- coef(model)[3]
> print(yGlucose)
   Glucose
```

| | |
|---|---|
| names | chr [1:3] "siri" "chiru" "loki" |
| num | 4L |
| num1 | 9L |
| num2 | 3L |
| num3 | 3 |
| Std | 1.58113883008419 |
| x | num [1:1000] -0.767 -0.816 -0.142 -0.278 0.436 ... |
| xBloodPressure | Named num 0.391 |
| y | num [1:1000] 0.299 -1.127 0.582 0.855 -0.384 ... |
| yGlucose | Named num 0.00451 |



RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

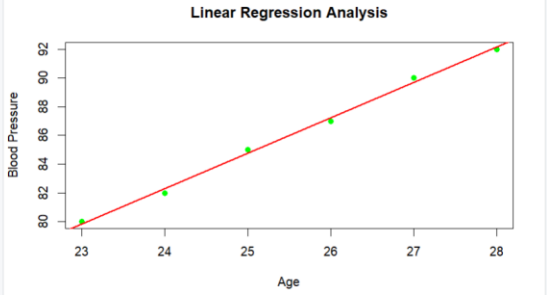
```
> # Create data frame
> diabetes <- data.frame(Age, BloodPressure, Glucose)
>
> # Multiple regression model
> model <- lm(Age ~ BloodPressure + Glucose,
+             data = diabetes)
>
> # Display model
> print(model)

Call:
lm(formula = Age ~ BloodPressure + Glucose, data = diabetes)

Coefficients:
  (Intercept)  BloodPressure        Glucose
    -8.672932       0.390977       0.004511

>
> # Display coefficients
> A <- coef(model)[1]
> print(A)
(Intercept)
  -8.672932
>
> xBloodPressure <- coef(model)[2]
> print(xBloodPressure)
BloodPressure
    0.3909774
>
> yGlucose <- coef(model)[3]
> print(yGlucose)
    Glucose
0.004511278
>
> # Regression equation
> cat("Age =", A, "+", xBloodPressure,
+     "* BloodPressure +", yGlucose, "* Glucose\n")
Age = -8.672932 + 0.3909774 * BloodPressure + 0.004511278 * Glucose
>
```

| Environment | |
|---|---|
| names | chr [1:3] "siri" "chiru" "loki" |
| num | 4L |
| num1 | 9L |
| num2 | 3L |
| num3 | 3 |
| Std | 1.58113883008419 |
| x | num [1:1000] -0.767 -0.816 -0.142 -0.278 0.436 ... |
| xBloodPressure | Named num 0.391 |
| y | num [1:1000] 0.299 -1.127 0.582 0.855 -0.384 ... |
| yGlucose | Named num 0.00451 |




28°C Partly sunny

Search

10:04 AM 26-10-2026